Evan Bruins

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EDUCATION

George Fox University, Newberg OR - B.S. *Mechanical Engineering* (GPA - 3.6)
AUGUST 2022 - PRESENT (GRADUATION MAY 2026)

Universitat Politècnica de València, València Spain - *Study Abroad*
JANUARY 2024 - MAY 2024
Immersed myself in Spanish culture while taking a full engineering course load.

RELEVANT EXPERIENCE

Analog Devices, Inc. - *Senior Design Capstone*
AUGUST 2025 - PRESENT

- Designed an automated laser depth measurement system for PVD targets to meet ±0.05 mm accuracy.
- Developed CAD models for a modular mechanical system using an 80/20 frame, motor driven turntable, and clamping mechanisms.
- Performed testing and analytical validation to eliminate mechanical leveling by software alignment.
- Presented the full system design to Analog Devices and received approval to proceed.

Intel, Portland OR - *Mechanical Design Intern*
AUGUST 2024 - NOVEMBER 2024

- Supported development of Intel's next generation Thermal Frame Loading Mechanism (TFLM) for CPU to socket mounting in server platforms.
- Designed test fixturing in Creo to measure axial shear forces of various thermal interface materials (TIMs).
- Developed and executed a custom testing procedure to collect data.
- Analyzed test results in Excel to inform design improvements.

PROJECTS

Custom RC Car - *Personal*
Independently designed and CAD modeled the full mechanical system in SolidWorks, completing multiple prototype iterations and packaging commercially sourced electronics for optimized performance.

Oscillating Air Engine - *GFU*
Designed and fabricated an air engine. Optimized for speed, and came first in that category.

FEA of a 3D Printed AFO - *GFU*
Conducted FEA within an engineering team to analyze a 3D printed ankle foot orthosis, validating SolidWorks Simulation models against experimental stiffness data and evaluating the effects of material properties, mesh selection, and boundary conditions on deformation and stress predictions.

Brayton Cycle Performance Study - *GFU*
Modeled a T55-GA-5512 gas turbine as a Brayton cycle to evaluate efficiency and power focused design modifications, including heat recovery and staged expansion, developing baseline, efficiency and horsepower optimized configurations.

SKILLS

SolidWorks (CSWA)
PTC Creo
FEA (SolidWorks Simulation)
Mechanical Design & Prototyping
3D Printing
MATLAB
Microsoft Excel
Multibody Dynamics
Experimental Validation
Spanish (Conversational)
German (Basic)

RELEVANT COURSES

Heat Transfer
Machine Dynamics and Vibrations
Fluid Dynamics
Design of Machine Elements
Controls

CLUBS

EMC - Engineering Missions Club

Formula Fox - RC racing club, founder and treasurer